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PROPERTIES OF LIQUID PbS, PbSe AND PbTe

N. M. RAVINDRA and V. K. SRIVASTAVA
Department of Physics, University of Roorkee, Roorkee-247672, India

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Abstract—We present here a study of the various properties of liquid PbS, PbSe and PbTe. In the first part of the presentation, we discuss the various possible models in the light of the semiconducting properties of these liquids. The electrical conductivity studies lead us to believe that the temperature dependence of electrical conductivity for these liquids is in accord with our earlier reported[1] studies of the temperature dependence of the energy gap for crystalline PbS, PbSe and PbTe. We also give an interpretation for σ_0, and discussion of the Hall coefficient and the Hall mobility is presented. The next part of the presentation deals with thermal conductivity, thermoelectric power and other related properties. The negative values of the thermoelectric power for these liquids are qualitatively analysed. Our studies of the magnetic susceptibility for crystalline and liquid PbTe indicate that the ratio of the transition term to the energy gap is a constant at all temperatures so that the Van-Vleck component is temperature independent, thereby explaining the constancy of the total susceptibility. Electronegativity and ionicity studies are presented. Finally, we report our studies of the density, viscosity and other physio-chemical properties.

NOMENCLATURE

a	lattice parameter
d	density
e	electronic charge
E_b	activation energy of viscous flow
E_G	energy gap
I	bond ionicity
I_c	effective ionicity
K	Thermal conductivity
k	Boltzmann constant
L_f	heat of fusion in liquid phase
P	pressure
Q_f	heat of fusion in crystalline phase
R	universal gas constant
R'	complexing ratio
R_H	Hall Coefficient
S	Seebeck coefficient or thermoelectric power
S_b	entropy of viscous flow
s_s	entropy per one electron in solid
s_l	entropy per one electron in liquid
s_f	entropy of fusion
T	temperature
T_{mp}	melting point
Z	average coordination number
β_s	Volume expansion coefficient
$-\Delta H_{298}$	heat of formation
Δx	electronegativity
μ_H	Hall mobility
μ_n	mobility of electrons
μ_p	mobility of holes
η	dynamic viscosity
ν	kinematic viscosity
π	Peltier coefficient
χ	magnetic susceptibility
σ	electrical conductivity

INTRODUCTORY SURVEY

Before we take up a study of the properties of liquid PbS, PbSe and PbTe, it would be worthwhile to look at the various possible models and classifications proposed by various workers.[4–10] In Tables 1 and 2, we summarise the classification of liquid semiconductors proposed by various workers.[4–10] Table 3 refers to the classification of liquid PbS, PbSe and PbTe and is based on Tables 1 and 2. PbS, PbSe and PbTe are

Table 1. Predominant characteristics of transport measurements on liquids[4]

Range	Number of liquids	Range of σ values (Ω^{-1} cm^{-1})	Sign of $d\sigma/dT$	Sign of S	Sign of $d\|S\|/dT$	R_H (cm^3/C)	Sign of $d\|R_H\|/dT$
A	38	10^5–5×10^3	− (34; 2)*	− (17; 10)	+ (19; 4)	$\sim 1/n_v N_a e$ (14; 3)	0 (15; 0)
B	34	5×10^3–10^2	+ (28; 5)	+ (14; 2)	− (13; 3)	$\sim 1/2N_a e$ (10; 3)	0 or + (8; 2)
C	39	10^2–10^{-12}	+ (39; 0)	+ (14; 2)	− (13; 1)	?	?

* First and second numbers in the parentheses are the number of times the predominant and a different characteristic were found.

Table 2. A summary of classification of liquid semiconductors by various workers

Workers	Classification	Comments
Regel[5]	Type I	Semiconductor → semiconductor transition on melting. $\sigma < \sim 10^4\,\Omega^{-1}$ cm^{-1}.
	Type II	Semiconductor → metal transition on melting. $\sigma > 10^4\,\Omega^{-1}$ cm^{-1}.
	Type III	Semiconductor → semiconductor → metal, the transition taking place at $T \gg T_{mp}$.
Glazov *et al.*[6,7]	Type IV	Semiconductor → ionic conductor (this happens in the case of cuprous iodide).
Mott[8]	Type a	Typical metals. $\sigma > 5000\,\Omega^{-1}$ cm^{-1}. $d\sigma/dT$ is usually negative. However $d\sigma/dT$ is positive in some divalent metals.
	Type b	Liquids of intermediate range. $\sigma = 300$–$5000\,\Omega^{-1}$ cm^{-1}: $d\sigma/dT$ is $+ve$.
	Type c	Semimetals or narrow gap semiconductors $\sigma = 1$–$300\,\Omega^{-1}$ cm^{-1}. $d\sigma/dT$ is positive. Hopping conduction and band conduction contribute to conductivity.
	Type d	Semiconductors. Hopping conductivity is negligible.
Enderby[9]	Category I	Electrical properties do not show irregular changes with composition.
	Category II	Electrical properties show irregular changes with composition.
Cutler[10]	Type M	Metals. Electrical properties are weakly dependent on temperature. $\sigma > 3000\,\Omega^{-1}$ cm^{-1}.
	Type S	Semiconductors. Electrical properties are strongly dependent on temperature. $\sigma \gtrsim 3000\,\Omega^{-1}$ cm^{-1}.

Table 3. Classification of liquid PbS, PbSe and PbTe by various workers

Workers	Classification PbS	PbSe	PbTe
Regel[5]	Type I	Type I	Type I
Mott[8]	c	b	b
	d	d	d
Allgaier[4]	B	B	B
		C	
Enderby[9]	II	II	II
Cutler[10]	S	S	S

semiconductors in the solid and liquid phase. They undergo a semiconductor → semiconductor transition on melting. Hence they may be classified as type I according to the definition of Regel.[5] Under Mott's classification,[8] liquid PbSe and PbTe may be regarded as type b since their electrical conductivities lie in the range 300–5000 Ω^{-1} cm^{-1} and liquid PbS may be regarded as type c since its conductivity is in the range 1–300 Ω^{-1} cm^{-1}. However, since PbS, PbSe and PbTe are essentially semiconductors, with the electrical conductivity obeying

$$\sigma_T = \sigma_0 \exp(-E_G/kT)$$

even in the liquid phase, as we shall see later, they may be classified as type d. According to the classification of Allgaier,[4] since PbS, PbSe and PbTe have conductivities in the range 100–5000 Ω^{-1} cm^{-1}, they may be classified as type B. It is to be noted however that the sign of the thermoelectric power is negative for these three lead compounds. Since what we are really looking for is σ in the range 5×10^3–100 Ω^{-1} cm^{-1} coupled with a positive $d\sigma/dT$ and a negative value of the thermoelectric power, it becomes difficult to classify liquid PbS, PbSe and PbTe under any of the types—A, B or C, as proposed by Allgaier. Based on the Hall mobility values, PbTe has also been classified under type C.[9] On the basis of a speculative approach, Enderby[9] and co-workers[11–13] assume that the Hall coefficient rather than the thermopower remains a useful measure of both the sign and the density of the current carriers. Since the electrical properties of PbS, PbSe and PbTe show irregular changes with composition, they may be classified under type II of the Enderby model. Liquid PbS, PbSe and PbTe have conductivities less than 3000 Ω^{-1} cm^{-1} and they undergo a semiconductor → semiconductor transition on heating. They may therefore be classified under type S of the Cutler's model.

ELECTRICAL CONDUCTIVITY

In this paper, we attempt to arrive at a qualitative understanding of the semiconductor → semiconductor transition in PbS, PbSe and PbTe.

Liquid PbS, PbSe and PbTe have conductivities very much greater than 1 Ω^{-1} cm^{-1}. Thus these liquids conduct electronically, as ionic conductivities are unlikely to be greater than $\sim 1\ \Omega^{-1}$ cm^{-1}. Also electrical conductivities of these liquids show a strong increase with increase in temperature, they are thus termed liquid semiconductors. Otherwise they are statistically degenerate and obey Fermi–Dirac statistics and in this sense, they are metals.

For those systems wherein there is a change from a semiconductor to metal on melting, a correlation between such a change and a decrease in the atomic volume exists.[2] It is well known that the crystalline semiconducting behaviour is associated with a large atomic volume and a small coordination number and for liquid PbS, PbSe and PbTe, this seems to be preserved in the liquid state also. It is interesting to note that most of the known liquid semiconductors contain a group VI B element.

Studies conducted recently[10] show that in some cases, the sensitivity of electrical conductivity to temperature cannot be ascribed to excitation of carriers across a gap or out of traps, but must reflect a change in chemical composition with temperature. Hence, the temperature dependence of electrical conductivity may be different from conventional semiconductors. Ioffe and Regel[2] have earlier emphasised on the fact that the mean freepath in semiconducting liquids is of the order of interatomic distance in contrast to liquid metals.

However, we find that for liquid PbS, PbSe and PbTe, the electrical conductivity is given by

$$\sigma_T = \sigma_0 \exp(-E_G/kT) \tag{1}$$

where the symbols have their usual meanings (see Nomenclature). In Fig. 1, we have a plot of the behaviour of σ on melting PbS, PbSe and PbTe. In Table 4, we also have some properties of PbS, PbSe and PbTe at the melting point. As can be seen in the table, the conductivity in the liquid phase σ_l is greater than σ_s, the conductivity in the solid phase for PbSe and PbTe, while for PbS, $\sigma_l < \sigma_s$. Of course, the change in the conductivity for PbS at its melting point is just about 12% $[(\sigma_s - \sigma_l)/\sigma_s] \times 100$. We note in the passing that the behaviour of σ with T for PbTe before and after melt are almost similar

$$\text{i.e. } (d\sigma/dT)_{\text{before melt}} \simeq (d\sigma/dT)_{\text{after melt}}.$$

This seems to reflect the possibility of the ordering being retained in the liquid phase. Thus assuming that the short range order of a solid is retained in the molten state, Mott[14] has earlier deduced a dependence which relates the value of the change in the

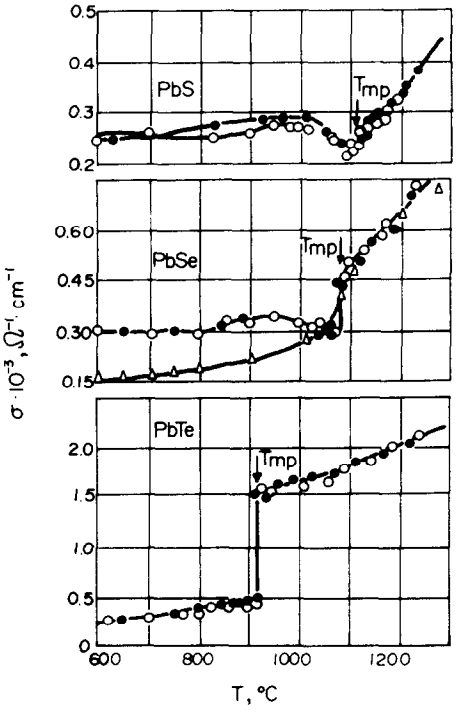


Fig. 1. Behaviour of σ on melting PbS, PbSe and PbTe.[3]

electrical conductivity at the melting point to the heat of fusion L_f. The relation is

$$(\sigma_s/\sigma_l) = -\exp(80\, L_f/T_{mp}). \tag{2}$$

The values of L_f are tabulated in table 4. The change in the actual conductivity of a semiconductor on fusion may be expressed by the formula

$$(\sigma_l/\sigma_s) = (\mu_l/\mu_s)\exp(-\Delta E/2k\, T_{mp}). \tag{3}$$

Now, looking at expression (1), the electrical conductivity is a function of the energy gap and temperature. But the energy gap itself is a function of temperature. Our earlier

Table 4. Properties of PbS, PbSe and PbTe at the melting point

Property	PbS	PbSe	PbTe
$\sigma_s 10^{-3}\,\Omega^{-1}\,cm^{-1}$	~0.25	~0.30	~0.42
$\sigma_l 10^{-3}\,\Omega^{-1}\,cm^{-1}$	~0.22	~0.45	~1.52
σ_s/σ_l	~1.14	~0.67	~0.28
L_f K·J/gm atom	−2.22	+6.9	+19.13
$S_s \mu V/K$	−220	−120	−60
$S_l \mu V/K$	−200	−60	−10
S_s/S_l	1.1	2.0	6.0
$\Delta S = \lvert S_s - S_l \rvert \mu V/K$	20	60	50
$\pi_s V$	−0.306	−0.163	−0.071
$\pi_l V$	−0.278	−0.082	−0.012
$\Delta\pi = \lvert \pi_s - \pi_l \rvert V$	0.028	0.081	0.059
$\Delta s = s_s - s_l\, 10^{23}$ J/K	0.25	0.98	0.80
d_s gm·cm^{-3}	7.07	7.57	7.69
d_l gm·cm^{-3}	6.45	7.10	7.45
$\Delta d = d_l - d_s$ gm·cm^{-3}	−0.62	−0.47	−0.24
$\Delta d(\%) = [(d_l - d_s)/d_s](\%)$	−8.8	−6.2	−3.1
(dT_{mp}/dP) K/at.	+0.0130	+0.0099	+0.0054
$\beta_s\, 10^5$ K^{-1}	4.8	5.7	6.4
S_f cal (gm atom·K)$^{-1}$	~3.1	~3.1	~3.1

studies[1] have shown that the temperature dependence of the energy gap for crystalline PbS, PbSe and PbTe is given by

$$E_{GT} = E_{G0} + \frac{\alpha T^2}{T + \beta} \tag{4}$$

Using equation (4) in equation (1), we have

$$\sigma_T = \sigma_0 \exp\left[- \frac{E_{G0}(T + \beta) + \alpha T^2}{kT(T + \beta)} \right] \tag{5}$$

where E_{G0} is the energy gap at 0 K, α and β are a measure of the electron phonon interaction and the Debye temperature respectively and are empirical constants. The values of α, β and E_{G0} are given in Table 6.

In this presentation, we find that expression (5) explains the temperature dependence of the electrical conductivity fairly well for liquid PbS, PbSe and PbTe. In Fig. 2, we have a plot of $\ln\sigma_T$ as a function of $1/T$ for these liquids. As can be seen in the figure, the agreement between experimental and calculated values is quite good. In Table 5, the extrapolated values of σ_0 are tabulated for the above liquids. It is to be noted that these calculations are based on the experimental data of Glazov *et al.*,[3] who proposed a similar expression to (5), except for the temperature dependence of the energy gap.

The fact that expression (5) is able to explain the temperature dependence of electrical conductivity for liquid PbS, PbSe and PbTe reflects the fact that the aspects of behaviour in fluidity are not the dominant elements in electrical behaviour, which has earlier been pointed out by Cutler.[10]

Since we are mainly concerned with semiconductors in the molten phase, we may presume that the system is nearly wholly degenerate with all the electrons contributing to

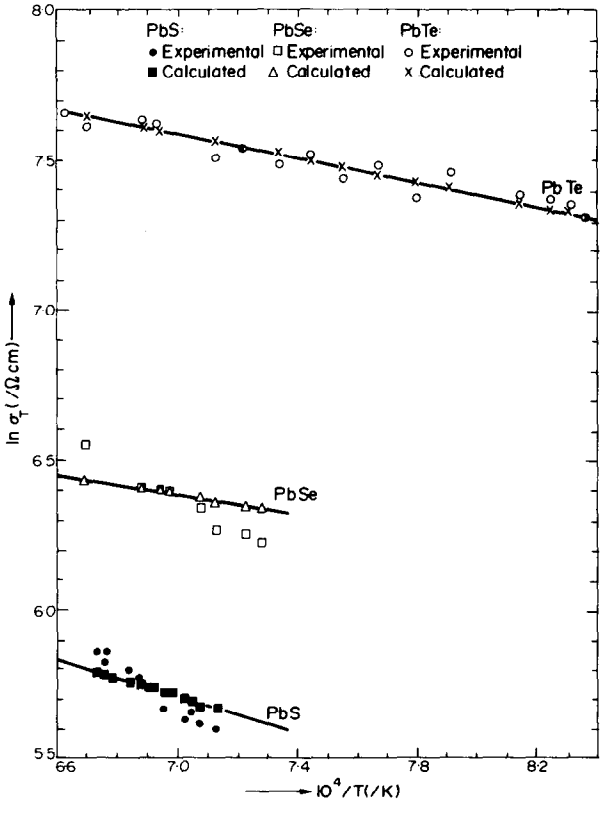


Fig. 2. Experimental and calculated values of $\ln\sigma_T$ as a function of $1/T$ for liquid PbS, PbSe and PbTe.

Table 5. Properties of liquid PbS, PbSe and PbTe

Property	PbS	PbSe	PbTe
$\sigma\ \Omega^{-1}\ cm^{-1}$	220	450	1520
Sign of $d\sigma/dT$	+	+	+
S $\mu V/K$	−200	−60	−10
Sign of dS/dT	+	+	−
$R_H\ cm^3\ C^{-1}$			-1.0×10^{-4}
$R_H/\lvert R_{H0}\rvert$			−2.1
$\mu_H\ cm^2\ V^{-1}\ Sec^{-1}$			0.15
$\sigma_0\ \Omega^{-1}\ cm^{-1}$	8.97×10^5	5.65×10^5	2.61×10^6
$\gamma\ 10^{-4}\ eV \cdot K^{-1}$	7.25	6.85	8.17
$S_0\ \mu V/K$	−313.567	−199.116	85.73
$m\ \mu V/K^2$	0.079	0.092	−0.0752
S'	10	7	1.9
$-A$ centistokes	0.595	2.854	1.869
B centistokes K	−13.64	2962	1496
Δv	0.0519	0.0334	0.0106
$d_{0l}\ gm \cdot cm^{-3}$	7.791	7.853	8.411
$-b_l x\ 10^3\ gm \cdot cm^{-3}\ K^{-1}$	0.958	0.575	0.843
d_l'	0.0234	0.0065	0.0148
Δx	0.9	0.8	0.5
l	0.183	0.148	0.061
l_c	−0.361	−0.420	−0.566
ΔT °C	~50	~50	~140
$E_b\ 10^{-3}$ cal/mol	9.80	6.85	6.85
$-S_b$ cal/mol·K	7.5	7.5	6.0
Sign of $d(d)/dT$	−	−	−
Sign of $d\chi/dT$			~0
Sign of $d\eta/dT$	−	−	−
Sign of dv/dT	−	−	−

conduction. Hence the conductivity is also given by the conductivity of a free electron gas. The expression is

$$\sigma = \frac{ne^2T}{m^*} \tag{6}$$

where n is the number of electrons per unit volume, e is the electronic charge, m^* is the effective mass and T is the time of relaxation. Because of the fact that the system is nearly wholly degenerate, n should be a constant independent of temperature. Therefore an increase in σ with T is now because of the temperature dependence of T and m^*. If L is

Table 6. Some properties of crystalline PbS, PbSe and PbTe*

Property	PbS	PbSe	PbTe
Structure	Galenite	Galenite	Galenite
Molecular weight a.m.u.	239.26	286.16	334.80
a Å	5.94	6.13	6.45
Z	6	6	6
T_{mp} K	1392	1361	1190
Q_f K·cal/gm atom	4.35	4.25	3.75
$-\Delta H_{298}$ K·cal/gm atom	11.25	10.5	8.4
E_{G0} eV	0.29	0.15	0.19
$\alpha\ 10^{-4}\ eV \cdot K^{-1}$	5	5	5
β K	40	40	40
$\mu_n\ 10^4\ cm^2/V \cdot sec$ (4.2 K)	80	14	80
$\mu_p\ 10^3\ cm^2/V \cdot sec$ (77.4 K)	15	14	17
$d_0\ gm \cdot cm^{-3}$	7.561	8.188	8.339
$-b \times 10^3\ gm \cdot cm^{-3}\ K^{-1}$	0.352	0.441	0.538
d'	0.0079	0.0062	0.0029
Sign of $d\sigma/dT$	+	+	+
Sign of $d(d)/dT$	−	−	−
Sign of dS/dT	−	−	−
Sign of $d\chi/dT$			~0

* Unless otherwise stated, the properties are all at 300 K.

the mean free path, v is the velocity of an electron at the Fermi surface and k_F is the wave number, then the relaxation time is given by

$$T = (L/v) = (m^*L/\hbar k_F) \tag{7}$$

and

$$n = (8\pi/3)(k_F/2\pi)^3 \tag{8}$$

n has earlier been seen to be temperature independent. That is, k_F, the wave number is a constant at high temperature. m^* is generally found to decrease with increase in temperature at high temperatures.[15] Using (7) in (6) and simplifying, we have

$$\sigma = ne^2L/\hbar k_F. \tag{9}$$

An increase in σ with T would therefore reflect an increase of L with T. Therefore based on a free electron approximation, one may conclude that the increase in electrical conductivity with temperature is because of an increase in the mean free path. This is consistant with the conclusion one may draw from the expression

$$\sigma = S_F e^2 L/12\pi^3\hbar \tag{10}$$

where S_F is the Fermi surface area equal to $4\pi k_F^2$.

We may also discuss the temperature dependence of conductivity in terms of mobility. After all, the electrical conductivity is given by

$$\sigma = ne\mu. \tag{11}$$

However, for a partially filled band, the conductivity may be given by[16]

$$\sigma = ne^- \bar{\mu}_n \quad \text{or} \quad \sigma = pe^+ \bar{\mu}_p \tag{12}$$

where n and p represent the number of electrons and holes respectively, e^-, e^+ represent the electron and hole charge respectively and $\bar{\mu}_n$, $\bar{\mu}_p$ are their respective average carrier mobilities. It has been pointed out by Allgaier[16] that for crystalline solids, it is more convenient and useful to employ the first or second form when the band is nearly empty or nearly full respectively. It is to be noted here that the first form is more fundamental. Irrespective of whether a material is n or p type, solid or liquid, crystalline or amorphous, σ, the electrical conductivity is always due to the electrons present. Based on the free electron approximation, we may therefore conclude that an increase in the electrical conductivity with temperature reflects an increase in the mobility of the charge carriers. Therefore for liquid PbS, PbSe and PbTe

$$d\sigma/dT = k(d\mu/dT) \tag{13}$$

where $k = ne$ and is temperature independent.

At this stage it would be worthwhile to qualitatively look at expressions (1) and (5). The total temperature dependence of the energy gap in semiconductors is given by the phenomenological expression

$$(dE_G/dT)_P = (\partial E_G/\partial T)_V + (\partial E_G/\partial T)_T \tag{14}$$

where the first term is due to phonons and the second one corresponds to the thermal expansion effect. If α' is the volume expansion coefficient and β' is the compressibility, then

$$(\partial E_G/\partial T)_T = (\alpha'/\beta')(\partial E_G/\partial P)_T \tag{15}$$

Since expression (4) explains the temperature dependence of the energy gap and hence the conductivity for liquid PbS, PbSe and PbTe, expressions (14) and (15) should also be applicable for these liquids. Incidentally, we wish to point to an interesting and systematic increase of the electron phonon contribution to the temperature dependence of the energy gap with increasing lattice parameter for crystalline PbS, PbSe and PbTe (see Fig. 3). As we shall see later, for these substances the interatomic spacing increases with

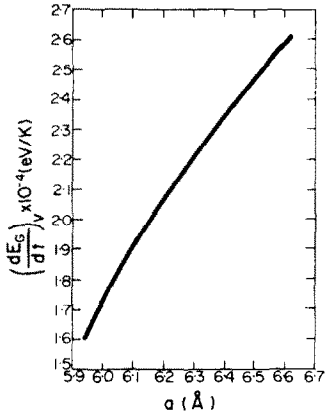


Fig. 3. $(dE_g/dT)_V$ vs lattice parameter for crystalline PbS, PbSe and PbTe.

increase in temperature not only in the solid phase but also in the liquid phase, thus reflecting a decrease in the density. It would thus be worthwhile to look at such a variation of the electron phonon contribution to the change in E_G with the lattice parameter for the liquid phase also.

INTERPRETATION FOR σ_0

It has been pointed out earlier[18] that hopping is important in liquid semiconductors only when $\sigma < 0.1\,\Omega^{-1}\,\mathrm{cm}^{-1}$. This justifies our classification of liquid PbS, PbSe and PbTe as type D under the classification suggested by Mott.[8] Based on the model of Mott and Davis,[19] σ_0 may be expressed as

$$\sigma_0 = C\exp(\gamma/k) \tag{16}$$

where

$$C = (0.06\,e^2/\hbar a_E) = (1.46 \times 10^{-5}/a_E) \tag{17}$$

$$\gamma \cong \mathrm{d}E_G/\mathrm{d}T$$

and a_E is the distance between localised states. C characterises the value of the conductivity when metallic conduction first appears. It is to be noted that this expression (17) for C is applicable for coordination number $Z = 6$. For other values of Z, it should be multiplied by $(6/Z)^2$. For PbS, PbSe and PbTe, $Z = 6$[3] (see Table 6). According to the model of Mott, we define the minimum metallic conductivity as $C = 200\,\Omega^{-1}\,\mathrm{cm}^{-1}$.[20] This would correspond to $a_E = 7.3$ Å. The calculated value of γ are presented in Table 5. γ is of the order of $2\mathrm{d}E_G/\mathrm{d}T$ for liquid PbS, PbSe and PbTe. In fact it has been observed that $\mathrm{d}E_G/\mathrm{d}T$ for liquid phase is greater than $\mathrm{d}E_G/\mathrm{d}T$ for solid phase.[10]

HALL MOBILITY

The Hall coefficient R_H is given by the expression

$$R_H = -1/ne \tag{18}$$

where n is the density of valence electrons and e is the electronic charge. The relation (18) which is essentially derived from the free electron model has been shown to be accurately obeyed for a large number of pure liquid metals.[21] In fact this relation (18) is obeyed better in liquids than in crystalline metals. This has been ascribed to the disorder in the liquid which results in a smooth Fermi surface.[10] It is this observation which has coloured the interpretation of subsequent investigation for liquid semiconductors. Generally speaking, the sign of R_H is found to be negative at all temperatures in contrast to the positive sign of S, the seeback coefficient or thermoelectric power.

However, as we shall see later in our presentation, for PbS, PbSe and PbTe, even the thermoelectric power is negative. Now, in case the transport or conductivity is by holes,

that is empty states in a nearly filled band, then according to the conventional transport theory, the sign of S and R_H is positive.

Relation (17) may be written as

$$\begin{aligned} R_H &= 1/pe \text{ for holes} \\ R_H &= -1/ne \text{ for electrons.} \end{aligned} \tag{19}$$

It has been pointed out earlier[10] that a possible explanation for the discrepancy in sign between R_H and S may be sought in terms of conduction in more than one band. However for liquid PbS, PbSe and PbTe no such discrepancy exists. In equation (18) n represents the density of free electrons and is usually calculated by assuming that all the valence electrons are free to take part in the process of electronic conduction. We may therefore conclude that R_H is essentially a constant for a given substance and is temperature independent.

The Hall mobility μ_H is given by

$$\mu_H = R_H \sigma. \tag{20}$$

For our system under study, since σ is increasing with T, we would expect μ_H also to increase with T. But looking at this problem of the temperature dependence of Hall mobility in terms of a two band model

$$\sigma = \sigma_n + \sigma_p$$

and

$$\mu_H = \frac{\mu_n \sigma_n + \mu_p \sigma_p}{\sigma_n + \sigma_p} \tag{21}$$

where μ_n and μ_p are both independent of T, σ_n and σ_p are functions of T and the ratio σ_n/σ_p increases with T. Relation (21) would however predict a decrease of μ_H with T which is in fact the case for Thallium telluride.[10] If we however consider μ_n and μ_p to be increasing with T, then we may easily explain an increase of μ_H with T. In Fig. 4, we have a plot of μ_H as a function of σ for various liquid semiconductors.[4,16] As can be seen in the figure, there is a systematic decrease in the Hall mobility with decrease in electrical conductivity. Of course, the plot is based on experimental data and is just a statistical one. It becomes quite difficult to conclude from such a plot and any sort of generalisation could be risky.

For a degenerate system such as liquid PbS, PbSe and PbTe, we would expect the Hall mobility to be the same as the conductivity mobility.

We wish to point out in the passing that PbS, PbSe and PbTe have very high values of the static dielectric constant.[17] Of course, for these systems, the static as also the high

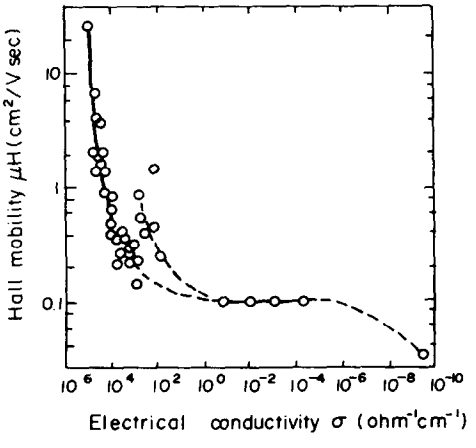


FIG. 4. Correlation between μ_H and σ in liquid semiconductors.[16]

frequency dielectric constants decrease with increase in temperature which is in consistency with the Moss relation.[22] Afterall, energy levels in a dielectric scale down by a factor inversely proportional to the square of the dielectric constant.[22] That is

$$E \propto \frac{1}{\epsilon^2} \tag{22}$$

Since E_G is increasing with T, we would expect ϵ to decrease with increase in T. Although the above discussion refers to crystalline lead compounds, since we find that E_G is increasing with T in the liquid phase also, we would expect such a correlation between E_G and ϵ to be very useful in the determination of dielectric constants for the liquid phase.

Our study seems to indicate that liquid PbS, PbSe and PbTe may essentially be considered as a highly doped semiconductor (n type). As we shall see later, our studies of the seebeck coefficient also supports this conclusion. For such a system, the potential energy should be that of a screened coulomb field and may be written as[19]

$$V(r) = -(e^2/Ke)\exp(-\gamma r). \tag{23}$$

It has been pointed out earlier that since PbS, PbSe and PbTe have very high values of the static dielectric constants, the potential $V(r)$ might be expected to be very weak and consequently the mobility might be high. This has been confirmed by Allgaier and coworkers[23,24] who report as high a value as $8 \times 10^5\ \mathrm{cm^2\ V^{-1}\ sec^{-1}}$ for mobility of these lead compounds (crystalline phase).

THERMAL CONDUCTIVITY

The total thermal conductivity is given by

$$K = K_a + K_e + K_r \tag{24}$$

where K_a is a contribution due to the atomic motion, K_e is due to electronic motion and K_r is due to radiation. If C is the heat capacity per unit volume and D is the thermal diffusivity, then

$$K_a = CD. \tag{25}$$

At the melting point, most of the thermal motion of a liquid is vibrational in character. Therefore applying the equipartition law, we have

$$C = 3k/(4\pi a'^3/3) \tag{26}$$

where k is the Boltzmann constant and a' is the atomic radius. Since we are mainly concerned with systems at high temperature, vibrational energy diffuses with a mean distance $\cong 2a'$, so that

$$D \simeq 4a'^2\nu' \tag{27}$$

where ν' is the vibrational frequency. Using 27 and 26 in 25 and simplifying, we have

$$K_a = (9/\pi)(k\nu'/a'). \tag{28}$$

For the lead compounds, K_a has a very small value (see Fig. 5) at the melting point reflecting the fact that a' and ν' do not vary much about the values 2 Å and $\sim 3 \times 10^{12}\ \mathrm{sec^{-1}}$ respectively. K_a has also been correlated to molecular weight, density, melting point etc.[25] A systematic decrease of $K_a (= K_L$ in Fig. 5) with an increase in the average atomic weight $\bar{A}$ has been observed for semiconducting liquids at their melting point (see Fig. 5). The lead compounds essentially have $\bar{A} > 100$ and their values of the atomic thermal conductivity are of the order of 1×10^{-3} cal/cm deg sec.

K_e, the thermal conductivity due to electronic motion is related to the electrical conductivity by the Wiedemann–Franz law. The relation is given by

$$K_e = W_0 \sigma T \tag{29}$$

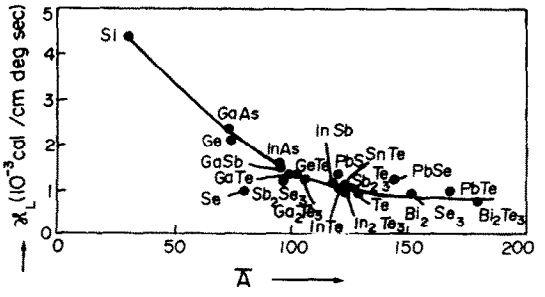


Fig. 5. The calculated atomic thermal conductivity as a function of the average atomic weight at the melting point of various liquids.[25]

where the constant $W_0 = (\pi^2/3)(k^2/e^2) = 2.45 \times 10^{-8}\,V^2/deg^2$. The Wiedemann–Franz law holds good for strongly degenerate systems. This metallic approximation can be expected to be accurate when $\sigma \gtrsim 500\,\Omega^{-1}\,cm^{-1}$ which would correspond to $K_e \gtrsim 3 \times 10^{-3}$ cal/deg·cm·sec at 1000 K. However for the lead compounds in the liquid phase, σ is close enough to this approximation. This apart, as we shall be seeing later, these compounds have a negative value of the seebeck coefficient reflecting the degeneracy. Expression (29) is therefore applicable for the lead compounds also. In Fig. 6, we present a plot of K_e as a function of T for liquid PbS, PbSe and PbTe.

For those systems wherein the electronic contribution to thermal conductivity dominates, the total thermal conductivity may be approximated by the Wiedemann–Franz law.

i.e. if K_a and $K_r \ll K_e$

then

$$K \simeq K_e = W_0 \sigma T. \tag{30}$$

If σ_b is the Stefan–Boltzmann constant, α is the absorption coefficient and n is the refractive index, then K_r, the thermal conductivity due to radiation is given by

$$K_r = (16/3)n^2(\sigma_b T^3/\alpha) \tag{31}$$

where the absorption coefficient α is small compared to the inverse length L^{-1} of the sample. K_r might appreciably contribute to heat transport for non metallic materials. For such materials, K_e would be relatively small. Based on the free electron approximation, we would expect the relation (30) to be valid for liquid PbS, PbSe and PbTe.

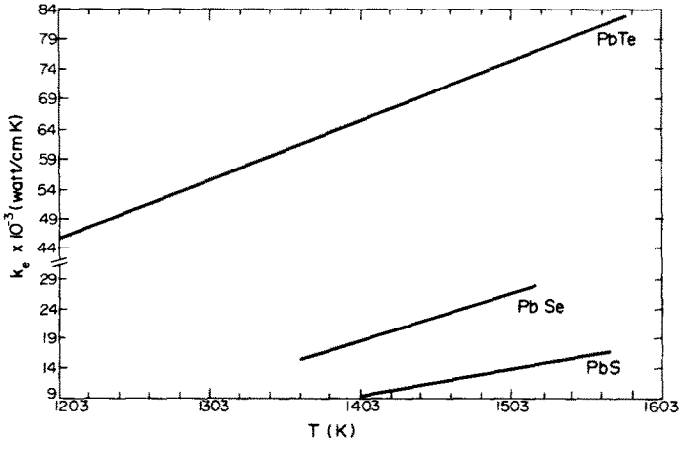


Fig. 6. Calculated values of K_e vs temperature for liquid PbS, PbSe and PbTe.

THERMOELECTRIC POWER OR SEEBECK COEFFICIENT

If ΔV is the voltage developed between two points of the material maintained at a small temperature difference ΔT, then the thermoelectric power or seebeck coefficient is given by $\Delta V/\Delta T$. Since we are basically dealing with n-type degenerate semiconductors, the thermoelectric power is given by

$$S = (k/e)\left[\frac{(E_c - E_F)}{kT} + A\right] \tag{32}$$

where e is negative, E_c is the energy of the conduction band edge and AkT is the average energy of the transported electrons measured with respect to E_c. A generally depends on the nature of the scattering process and is a constant between 2 and 4. If the current carrying mechanism is by holes, then the sign of S is reversed and $(E_c - E_F)$ will essentially be $(E_F - E_V)$. If we have ambipolar conduction in the system, then the thermopower associated with each carrier is weighted according to the contribution each makes to the total current. Thus for ambipolar transport, there is cancellation between the positive and negative contributions to S due to hole and electron transport respectively. Therefore for ambipolar conduction, the seebeck coefficient is given by

$$S = \frac{\sigma_p S_p + \sigma_n S_n}{\sigma_p + \sigma_n} \tag{33}$$

where

$$\sigma = \sigma_p + \sigma_n$$

σ_p, σ_n, S_p, S_n are respectively the holes and electrons conductivities and the hole and electron thermopowers.

Generally speaking, for molten semiconductor compounds, two essential features seem to be valid:

(i) The magnitudes of S and σ are strongly correlated. When σ is in the metallic range ($\sigma \gtrsim 10{,}000\,\Omega^{-1}\,\mathrm{cm}^{-1}$), S is generally small in magnitude ($\ll 100\,\mu$V/deg). When σ is smaller, the magnitude of S systematically increases into the range $\gtrsim 100\,\mu$V/deg.

(ii) Among liquids not in the metallic range of σ, S is frequently positive.

For liquid PbS, PbSe and PbTe, as we have seen earlier, the electrical conductivity is in the semiconducting range. But the thermopower S for all of them are negative reflecting the dominant role of electrons in their contribution to the seebeck coefficient. Holes contribute less to seebeck coefficient.

For liquid PbS, there is a sign inversion of S (see Fig. 7) from positive to negative. The positive value of S may be because of a dominant role of holes. With increase in T, this contribution is decreasing and hence the change in sign. An analysis of the changes in the seebeck coefficient at the melting point along with the data on changes in the density and electrical conductivity can yield additional information on changes in the short range order and in the nature of the chemical binding. And this apart, as we have seen, the sign of thermoelectric power may be used to determine the type of conduction.

Most of the detailed studies of liquid semiconductors have been done by Mott.[8,20,26,27] His model has been explained in detail by Cohen and coworkers[29] as also by Davies and Mott.[28] The essential features of his model are:

(a) E_V and E_C separate the extended from the localised states.

(b) Energy range of localised states is somewhat greater in the conduction band than in the valence band, thereby explaining the tendency for liquid semiconductors to exhibit positive thermoelectric powers. This model holds good for liquids having a composition range for which $\sigma \leq 200\,\Omega^{-1}\,\mathrm{cm}^{-1}$. For such materials

$$S = \frac{\pi^2}{3}\frac{k^2 T}{e}\left[\frac{d(\ln \sigma)}{dE}\right]_{E=E_F} \tag{34}$$

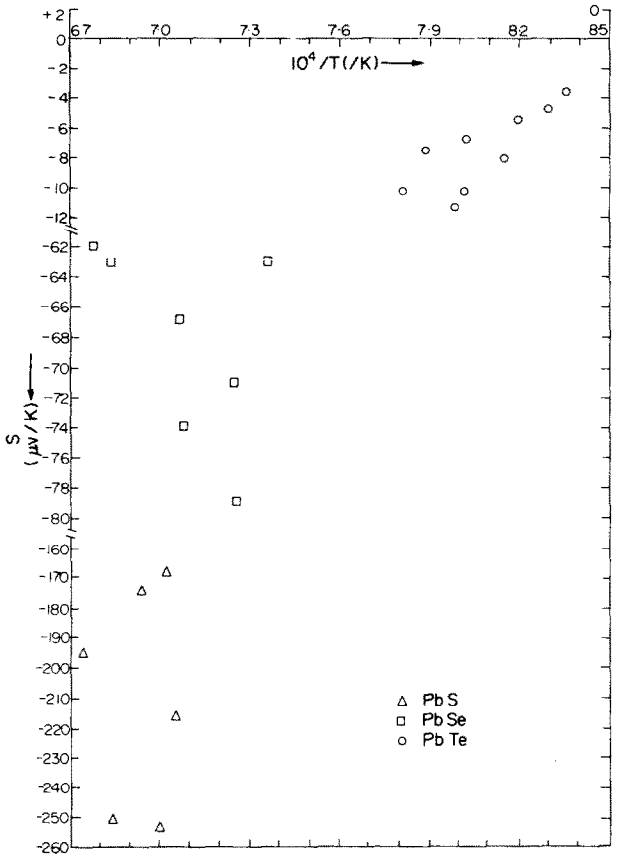


Fig. 7. Thermo electric power S as a function of $1/T$ for liquid PbS, PbSe and PbTe.

and

$$\sigma = D\exp(-E/kT) \tag{35}$$

where D is a constant and E is the characteristic energy. This characteristic energy has been identified by Mott to be

$$E_F - E_V = E - \gamma T \text{(holes)}$$

or

$$E_C - E_F = E - \gamma T \text{(electrons)} \tag{36}$$

Using (36) in (35), we have

$$\sigma = C\exp(\gamma/k)\exp(-E/kT) \tag{37}$$

where $C \sim 200\,\Omega^{-1}\,\text{cm}^{-1}$. From the work of Cutler and Mott,[30] we have that provided the current is carried by electrons (holes) excited to E_C (E_V), then

$$\begin{aligned} S_p &= \frac{k}{e}\left[\frac{E_F - E_V}{kT} + 1 - \frac{\gamma}{k}\right] \\ S_n &= \frac{k}{e}\left[\frac{E_C - E_F}{kT} + 1 - \frac{\gamma}{k}\right] \end{aligned} \tag{38}$$

Therefore, a plot of S as a function of $1/T$ should be a straight line with an intercept of $(k - \gamma)/e$.

After having had a look at the model, let us now apply the model to the case of liquid

PbS, PbSe and PbTe. These liquid semiconductors have negative values of the thermoelectric power. Hence based on the Mott's model, we may conclude that energy range of localised states is somewhat greater in the valence band than in the conduction band, thereby explaining the tendency of these lead compounds to exhibit negative thermoelectric powers. That is, if S is as given in expression (34), then $(\mathrm{d}\ \ln\sigma/\mathrm{d}E)_{E=E_F}$ should be a negative quantity.

Although the Mott's model is essentially applicable for liquids with $\sigma \leq 200\,\Omega^{-1}\,\mathrm{cm}^{-1}$, we tried this model for the lead compounds. In Fig. 7 we have a plot of S as a function of $1/T$ (based on the data of Glazov *et al.*[3]). As can be seen in Fig. 7, the plot cannot be equated to that of a straight line and hence the question of finding out the intercept does not arise. This seems to reflect the break down of Mott's model for liquid PbS, PbSe and PbTe. It is interesting to note here that for PbS, the change in σ and S on melting is insignificant (see Table 4). Such a change is a maxima for PbTe. With a knowledge of seebeck coefficient, it becomes trivial to calculate the Peltier coefficient. After all the two are related by the well known Kelvin relation

$$\pi = ST. \tag{39}$$

In general, the seebeck coefficient decreases with increase in temperature. However in certain temperature regions, there is some random variation.

FIGURE OF MERIT

The electrical conductivity and hence the energy gap is the dominant property of any material while determining its application for infrared detection, photovoltaic or any other device aspect. Pbs, PbSe and PbTe have their gaps in the infrared region and hence are very useful for infrared detection. In this study, we essentially find that the electrical conductivity and hence the energy gap does have a dominating role to play in thermoelectric conversion also.

After all, the thermoelectric efficiency of a semiconductor or the figure of merit Z' determines the potential value of liquid semiconductors in devices for thermoelectric conversion. Z' is given by

$$Z' = \frac{S^2\sigma T}{K} \tag{40}$$

where the symbols have their usual meanings. If $Z' \sim 1$, then considerable fraction of the Carnot efficiency can be derived from the thermoelectric conversion process.[31] In the light of our consideration of liquid PbS, PbSe and PbTe in terms of the free electron model, we have

$$K \cong K_e$$

Therefore using the Wiedemann–Franz law, we have

$$Z' \simeq S^2/W_0$$

for

$$S = \sqrt{W_0}, \quad Z = 1$$

With decrease in temperature, K_e decreases and S increases. But σ decreases. Thus Z' is a highly temperature sensitive quantity. This coupled with engineering and metallurgical problems in using liquids in high temperature devices has brought in practical limitations on application of liquid semiconductors for thermoelectric conversion.

MAGNETIC SUSCEPTIBILITY

The magnetic susceptibility for liquid metals is given by

$$\chi = \chi_{\mathrm{ion}} + \chi_P + \chi_L \tag{41}$$

where χ is the susceptibility, χ_{ion} is the ionic contribution, χ_P is the Pauli paramagnetic

term and χ_L is the Landau diamagnetic term. According to the free electron gas model,

$$\chi_P = -3\chi_L = \mu_B^2 N(E_F)$$

where μ_B is the Bohr magneton. χ_P and χ_L behave somewhat differently when the free electron model is refined to include the band structure and many body effects.

Since liquid PbS, PbSe and PbTe are essentially semiconductors, generally speaking we would expect them to be diamagnetic. Investigations of magnetic suscpetibility of PbTe have indicated that PbTe is diamagnetic both in the solid and liquid states. In Fig. 8, we have a plot of temperature dependence of magnetic susceptibility for PbTe in the solid and liquid phases. As can be seen in the figure, the magnetic susceptibility of PbTe varies slightly with temperature in the solid phase. In the liquid phase however, χ is practically independent of T. It is worth noting here that $(d\chi/dT)_{solid} - (d\chi/dT)_{liquid} \cong 0$ for PbTe. Such an observation is seen for σ also.

In general

$$\chi_T = \chi_d + \chi_{VV} \tag{42}$$

where χ_T is the total magnetic susceptibility, χ_d is the diamagnetic part and χ_{VV} is the Van-Vleck contribution. The diamagnetic susceptibility is independent of temperature. Therefore any change in χ_T with T is due to change in χ_{VV} alone. χ_{VV} may be written as

$$\chi_{VV} = 2N\mu_B^2 \sum_K \frac{|\langle l|L_Z|K\rangle|^2}{E_K - E_L} \tag{43}$$

where N is the Avogadro number, μ_B is the Bohr magneton and L_Z is the Z component of the orbital angular momentum which couples the ground state $\langle l|$ with energy E_l to the excited state $|K\rangle$ with energy E_K.

The matrix elements in equation (43) arise from virtual magnetic dipole transitions between states on the same site (hereafter numerator of equation (43) shall be referred to as the transition term). The temperature dependence of χ_{VV} is due to:

(a) Change in the transition term with temperature.
(b) Change in $E_K - E_l$ with temperature.

For the lead compounds under consideration, $E_K - E_l = E_G$, the energy gap of the semiconductor. For liquid PbTe, since χ_T ($=\chi$ in Fig. 8) is essentially temperature independent, the ratio of the transition term to the energy gap should be a constant at all temperatures. That is χ_{VV} should be constant thereby explaining the constancy of χ_T. Since the change in E_G with T is known for liquid PbTe, one may also evaluate the change in transition term.

The change in the magnetic susceptibility at the melting point could be due to an increase in the carrier density, making an additional contribution to the paramagnetic component of the susceptibility because of the spin paramagnetism or it could be due to an enhancement of the Van Vleck polarization paramagnetism, the role of which may be important.[32] The Van Vleck paramagnetic contribution to χ for a molecule can be described as the effect of mixing of higher states under the perturbation of a magnetic field.

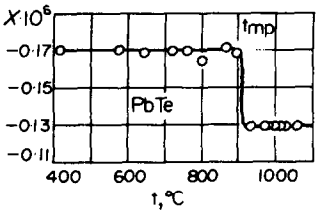


Fig. 8. Temperature dependence of the magnetic susceptibility of lead Telluride in the solid and liquid states.[3]

Semiconducting properties are due to covalent or covalent-ionic bonds between atoms. Consequently, the liquid PbS, PbSe and PbTe in the liquid phase have mainly covalent bonds. In the solid phase, covalent crystals have three dimensional networks of rigid bonds. Such networks do not exist in the liquid phase but are replaced by one dimensional molecular or chain-like structures. Therefore we may conclude that one dimensional structures may be formed in liquid PbS, PbSe and PbTe. The nature of the binding or packing of structural units in the liquid phase may be such that the short range order is still close to the solid state structure either over a fairly wide range of temperature or at least near the melting point. In the latter case, melting should result in the rearrangement of the short range order so as to produce a stable configuration which does not vary greatly when the temperature is increased. A decrease in the electrical conductivity of PbS at its melting point suggests that this compound has a molecular structure in the liquid state.

For liquid Te and In_2Te_3, χ is seen to increase with T above the melting point in parallel with a similar behaviour of $\sigma(T)$. Based on this, χ and σ are essentially seen to obey a relation of the type

$$\chi = A + B\sqrt{\sigma} \tag{44}$$

where A and B are constants. This result has been tested for Te and Tl–Te alloys. The relation (44) is seen to be obeyed in the diffusive region of σ ($\sigma \gtrsim 2000\,\Omega^{-1}\,cm^{-1}$). For liquid PbTe, although σ is essentially less than $2000\,\Omega^{-1}\,cm^{-1}$, a plot of χ as a function of $\sqrt{\sigma}$ would be a straight line with slope equal to zero and intercept equal to χ. This is because, although σ is varying with T, χ is independent of T for liquid PbTe (see Fig. 8).

The constant A in expression (44) has earlier been attributed to χ_{LL}, the diamagnetic susceptibility of both ions and molecules which is expressed theoretically by the Langevin–Larmor formula

$$\chi_{LL} = -(e^2/6mc^2)\sum_i \langle r_i^2 \rangle \tag{45}$$

where $\langle r_i^2 \rangle$ is the mean square distance of a particular electron from the centre of the molecule. However for liquid PbTe, such a plot of χ vs $\sqrt{\sigma}$ would not be informative as χ is independent of σ.

ELECTRONEGATIVITY AND IONICITY

If A_bB_a corresponds to a compound in which A and B exhibit common chemical valencies a and b respectively, when A is generally more electro positive constituent, then the difference in the electronegativity ΔX may be defined as[33]

$$\Delta X = X_B - X_A. \tag{46}$$

A general guide to the ionicity of bonds is provided empirically through ΔX. There have been various definitions of electronegativity. But generally, electronegativity is defined as a measure of the strength of attraction of an element for a valence electron. In Table 5, we list the values of ΔX for liquid PbS, PbSe and PbTe. Considering Pb as cations and S, Se and Te as anions, we find that on the replacement of an anion by a heavier ion, the electronegativity essentially decreases.

If P is the dipole moment of the bond and d is the bond distance, then the bond ionicity I is given by P/ed. Pauling has proposed an empirical formula relating I and ΔX. The relation is given by

$$I = 1 - exp(-\Delta X^2/4). \tag{47}$$

The effective ionicity I_c is given by[34]

$$I_c = 1 - [(1 - I)/R'] \tag{48}$$

where R' is defined as a complexing ratio and is the ratio of the coordination number of

an atom to its valence. For liquid PbS, PbSe and PbTe, $R' = 0.6$ (coordination number = 6, valency = 10). In Table 5, we also present the calculated values of I and I_c.

DENSITY

In Tables 4, 5 and 6, we list the other properties of interest to us like density, viscosity etc. As can be seen in the table, there is a maximum change in density on melting PbS, while for PbTe such a change is a minimum. It is interesting to note here that for PbS, while the difference in σ and S in the solid and liquid phase is a minima, the change in the density is a maxima. That is, the sudden change in the density at the melting point is reduced when anions are replaced with heavier elements in the series PbS → PbSe → PbTe. It has been concluded that the decrease in the density at the melting point especially if it is small in magnitude, can be regarded as an indication that the general nature of the chemical binding and the short range order is retained in the liquid state. We have earlier seen that such a conclusion may also be drawn on investigating the electrical conductivity.

A change in density after all reflects the change in atomic volume. The atomic volume and the nature of the binding are always correlated. Based on such a correlation, it has been concluded that many liquid semiconductors have covalent structure. Such a conclusion is based on the fact that covalent bonds usually require a relatively small coordination number Z, that is the number of nearest neighbours which is typically 4 or 6. We have earlier seen that $Z = 6$ for liquid PbS, PbSe and PbTe reflecting the fact that they essentially have a covalent structure. For a system which has metallic binding, the coordination number is of the order of 12 in solids and 9 in liquids reflecting the geometry of closely packed spheres in crystals and randomly packed spheres in disordered materials.

Unlike Ge, Si and some III–V semiconducting compounds which show an increase in the density upon melting, liquid PbS, PbSe and PbTe show a decrease in the density on melting. An increase in the density with temperature has earlier been attributed to an increase in the value of the coordination number Z thus permitting the system to undergo a semiconductor → metal transition. But generally on heating, for any system, the interatomic spacing should increase—thus resulting in an increase in the volume. This effect would tend to decrease the density. Therefore the total change in the density with increase in temperature for any system is essentially due to: (i), an increase in the coordination number, and (ii) an increase in the interatomic distance.

If the former effect dominates over the latter, we may expect the density to increase permitting a semiconductor → metal transition on melting. However for a converse case, the density would decrease with increase in temperature permitting no such semiconductor → metal transition.

Based on the above study, we might conclude that for liquid PbS, PbSe and PbTe the volume component dominates over Z thereby explaining the decrease in the density with increase in temperature. However it is of interest here to note that for these liquids, the electrical conductivity is increasing with increase in temperature permitting more and more electrons to contribute to conductivity thus tending towards a metallic or a free electron approximation while the decrease in density seems to preserve its semiconducting behaviour. This is probably characteristic of the lead compounds. Such an observation is true for Sn-Te and Ge-Te also.

VISCOSITY

The kinematic viscosity v is defined by the following equation

$$\eta = vd. \tag{49}$$

The units of η used in this presentation is the poise ($gm \cdot cm^{-1}\ sec^{-1}$) and v is in stokes (cm^2/sec). The kinematic viscosity is useful in an evaluation of the free activation energy of viscous flow (E_b). Glazov *et al.*[3] have reported their calculations of E_b. It is found that at $T > T_{melt}$, E_b increases linearly with T. This temperature interval (measured from the

melting point) above which such a linear dependence of E_b is observed is seen to increase along the series PbS → PbSe → PbTe, i.e. when an anion is replaced with a heavier element (see Table 5) (temperature interval = ΔT). The dynamic viscosity and hence the kinematic viscosity decrease with increase in temperature. This is essentially so because as has been observed earlier, the density decreases with increase in T.

The Frenkel's equation relates the dynamic viscosity to the activation energy of viscous flow. The relation is given by

$$\eta = \exp(E_b/RT) \tag{50}$$

where R is the universal gas constant and the rest of the symbols have their usual meanings. In Table 5, we present the value of the activation energy of viscous flow. Glazov *et al.*[3] have earlier analysed these properties for the lead compounds. The reader may refer to Glazov *et al.*[3] for detailed studies.

PRESSURE DEPENDENCE OF THE MELTING POINT

If Q_f is the heat of fusion and S_f, the entropy of fusion, then we have that

$$S_f = \frac{Q_f 10^3}{T_{mp}} \tag{51}$$

where T_{mp} is the melting point in K, Q_f is expressed in K·cal/gm atom, and S_f is in K·cal/gm deg atom.

The entropy of fusion S_f is a measure of the change in the degree of ordering after melting, due to changes in the structure and in the nature of the chemical binding. A high value of the entropy of fusion reflects considerable changes at the melting point. For liquid PbS, PbSe and PbTe, it is interesting to note that S_f is of the same order of magnitude. This leads us to conclude that similar changes take place in all the three compounds on melting. They retain their chemical binding even after melting.

From the Clausius Clayperon equation, one may evaluate the pressure dependence of the melting point. It is given by

$$(dT_{mp}/dP) = (V_l - V_s)/S_f = (T_{mp}/Q_f)(1/d_l - 1/d_s) \tag{52}$$

where $V_l - V_s$ represents the volume change at the melting point. For liquid Si, Ge and other III–V compounds, dT_{mp}/dP has a negative value. This is so because for these materials, $V_l - V_s$ is a negative quantity i.e., there is a decrease in the volume after melting or the interatomic spacing decreases. For the lead compounds, dT_{mp}/dP has a positive value. As we have seen earlier, the density decreases with increase in temperature. This has been attributed to an increase in the volume on melting explaining thereby positive dT_{mp}/dP. This value of dT_{mp}/dP decreases along the series PbS → PbSe → PbTe, i.e. when an anion is replaced by a heavier element.

BEFORE AND AFTER THE MELTING POINT

In this study, we will essentially point out to the temperature dependence of the various properties.

(i) The electrical conductivity obeys the exponential law both in the solid and the liquid phase. The temperature dependence of the energy gap is seen to obey our earlier reported relation.

(ii) The magnitude of the thermoelectric power which essentially increases with T for crystalline PbS, PbSe and PbTe varies linearly with T for the liquid phase

$$\text{i.e. } S = S_0 + mT$$

and S' is the r.m.s. error. m is the slope and T is in K (see Table 5).

(iii) The kinematic viscosity for the liquid phase obeys the relation,

$$\text{Log } \nu = A + B/T$$

where ν is the viscosity in centistokes, T is in K and $\Delta\nu$ is the r.m.s. error.

(iv) The density decreases linearly both for the solid and the liquid phase. For solid phase

$$d_s = d_0 + bT$$

where d_s is the density of the solid in gm/cm^3, d' is the r.m.s. error and the rest of the symbols have their usual meanings (see Table 6).

For the liquid phase

$$d_l = d_{0l} + b_l T$$

where d_l is the density of the liquid and d_l' is the r.m.s. error.

CONCLUSION

A study of the various properties of liquid PbS, PbSe and PbTe has been presented above. The first part of the study deals with a survey of the various models for liquid semiconductors. Our earlier studies of the temperature dependence of the energy gap for crystalline PbS, PbSe and PbTe have been shown to be valid for the liquid phase also. An interpretation for σ_0 has been given in terms of the Mott's model. Based on a free electron approximation, the Hall mobility, Hall coefficient, thermal conductivity, thermoelectric power and other related properties have been discussed. The Mott's model does not seem to explain the variation of the thermoelectric power with the reciprocal of temperature. Our studies of the temperature dependence of magnetic susceptibility for PbTe lead us to conclude that the Van Vleck component is temperature independent both for the crystalline and liquid phase. Also included are electronegativity, ionicity, density, viscosity and other properties.

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REFERENCES

1. RAVINDRA N. M., S. AULUCK & V. K. SRIVASTAVA, *Phys. Status Solidi* **a52**, K 151 (1979).
2. IOFFE A. F. & A. R. REGEL, *Progress in Semiconductors* (edited by GIBSON A. F.), Vol. 4, p. 237. Heywood. London (1960).
3. GLAZOV V. M., S. N. CHIZHEVSKAYA & N. N. GLAGOLEVA, *Liquid Semiconductors*. Plenum, New York (1969).
4. ALLGAIER R. S., *Phys. Rev.* **185**, 227 (1969).
5. REGEL A. R., Doctoral Dissertation, Lenningrad (1956).
6. GLAZOV V. M. & S. N. CHIZHEVSKAYA, *Zh. neorg. khim.* **9**, 759 (1964).
7. CHIZHEVSKAYA S. N., Dissertation for Candidate's degree, A. A. Baikov Metallurgy Institute (1963).
8. MOTT N. F., *Phil. Mag.* **24**, 1 (1971).
9. ENDERBY J. E., *Amorphous and Liquid Semiconductors* (edited by TAUC J.), p. 361. Plenum, New York (1974).
10. CUTLER M., *Liquid Semiconductors*. Academic Press, New York (1977).
11. ENDERBY J. E. & L. WALSH, *Phil. Mag.* **14**, 991 (1966).
12. ENDERBY J. E. & C. J. SIMMONS, *Phil. Mag.* **20**, 125 (1969).
13. ENDERBY J. E. & E. W. COLLINGS, *J. non-crystalline Solids.* **4**, 161 (1970).
14. MOTT N. F., *Proc. Roy. Soc.* **A146**, 465 (1934).
15. VASILEFF H. D., *Phys. Rev.* **105**, 441 (1957).
16. ALLGAIER R. S., *Phys. Rev.* **B2**, 2257 (1970).
17. RAVINDRA N. M. & V. K. SRIVASTAVA, *Phys. Status Solidi* **a57** (1980).
18. ANDREEV A. A., *Proc. 5th Int. Conf. Amorphous and Liquid Semiconductors*, p. 343. Garmisch-Partenkirchen (1973).
19. MOTT N. F. & E. A. DAVIS, *Electronic Processes in Non Crystalline Materials*. Clarendon Press, Oxford (1971).
20. MOTT N. F., *Phil. Mag.* **22**, 1 (1970).
21. FABER T. E., *Introduction to Theory of Liquid Metals*. Cambridge University Press, London (1972).
22. MOSS T. S., *Proc. phys. Soc.* **B63**, 167 (1950); *Photoconductivity in the Elements*, pp. 61-63, Butterworths, London (1952).
23. ALLGAIER R. S. & B. B. HOUSTON, *Proc. Int. Conf. Phys. Semiconductors*, Exeter, p. 172. The Institute of Physics and Physical Society, London (1962).
24. ALLGAIER R. S. & W. W. SCANLON, *Phys. Rev.* **111**, 1029 (1958).
25. REGEL A. R., I. A. SMIRNOV & E. V. SHADRICHEV, *Phys. Status Solidi.* **5**, 13 (1971).
26. MOTT N. F., *Phil. Mag.* **13**, 989 (1966).
27. MOTT N. F., *Adv. Phys.* **16**, 49 (1967).
28. DAVIES E. A. & N. F. MOTT, *Phil. Mag.* **22**, 903 (1970).
29. COHEN M. H., H. FRITZCHE & S. R. OVSHINSKY, *Phys. Rev. Lett.* **22**, 1065 (1969).
30. CUTLER M. & N. F. MOTT, *Phys. Rev.* **181**, 1369 (1969).

31. IOFEE A. F., *Semiconductor Thermoelements and Thermoelectric Cooling*. Infosearch, London (1957).
32. DORFMAN Ya. G., *Diamagnetism and Chemical Binding*. Fizmatgiz, Moscow (1961).
33. PAULING L., *The Nature of the Chemical Bond*, 3rd ed. Cornell University Press, New York (1960).
34. PHILLIPS J. C., *Bonds and Bands in Semiconductors*. Academic Press, New York (1973).